Pandemic Simulation Using Agent-Based Modeling and WebGL Visualization with Three.js

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***Abstract*—In this paper, we present a web-based pandemic simulation model utilizing agent-based modeling and Three.js, a powerful JavaScript 3D graphics library. The simulation provides a visual and interactive platform to observe how an infectious disease spreads among individuals within a population. By modeling individuals as agents with attributes such as position, state (healthy, infected, recovered), and movement behavior, the simulation replicates realistic infection dynamics. Our goal is to offer an intuitive tool for understanding how various parameters like infection rate, population density, and recovery time influence the spread. This system runs entirely in the browser, making it accessible for educational and exploratory use without specialized software. The paper details the simulation's methodology, system design, implementation, and observations, highlighting the importance of visual simulations in enhancing public understanding of epidemic behavior.**

***Index Terms*—Agent-Based Modeling, Three.js, WebGL, Pandemic Simulation, JavaScript, Visualization**

I. Introduction

Pandemics such as COVID-19 have highlighted the importance of understanding disease transmission and the dynamics of infection spread. Simulation models provide a powerful way to study and visualize these dynamics. Traditional models often rely on statistical methods or mathematical equations that may be less intuitive for non-experts. In contrast, agent-based models simulate the actions and interactions of autonomous agents, offering a more relatable and visual perspective. This paper introduces a browser-based simulation using Three.js that demonstrates how infection spreads through a moving population, aiming to enhance accessibility, learning, and experimentation.

II. Related Work

Several pandemic simulation tools and models have emerged, ranging from academic simulators like CovidSim to educational animations and dashboards. Tools such as NetLogo have long supported agent-based modeling, but often require installation or learning a custom language. Our simulation differs by being web-native, interactive, and visually engaging through the use of Three.js. Prior studies have shown that visual simulations can improve public comprehension and engagement with scientific topics, making this approach both technically and socially relevant.

III. System Design / Methodology

The simulation uses agent-based modeling, where each person in the population is represented by a sphere (agent) with individual state logic. States include:

- Healthy (blue)
- Infected (red)
- Recovered (green)

Agents move randomly within a bounded 3D space. When a healthy agent comes within a certain distance (infection radius) of an infected agent, it has a probability of becoming infected. After a predefined recovery time, infected agents transition to the recovered state.

Simulation parameters such as infection rate, population size, and movement speed are adjustable, enabling users to experiment and observe changes dynamically.

IV. Implementation & Technologies Used

The core technology used is Three.js, a WebGL-based 3D rendering library for JavaScript. Other supporting technologies include:

- HTML/CSS: For structure and minimal UI controls
- JavaScript: For simulation logic and state management

The system architecture includes:

- A scene with 3D agents rendered as colored spheres
- A loop that updates agent positions and handles state transitions
- UI sliders or input fields to adjust parameters in real time

Performance is optimized by limiting the agent count and using efficient collision detection mechanisms.

V. Results and Observations

The simulation successfully demonstrates how disease spreads in a population and how parameters like infection radius and agent density affect the outbreak curve. Observations include:

- Higher population density leads to faster and wider infection spread
- Lower infection probability significantly delays outbreak
- Recovered agents prevent reinfection, simulating herd immunity

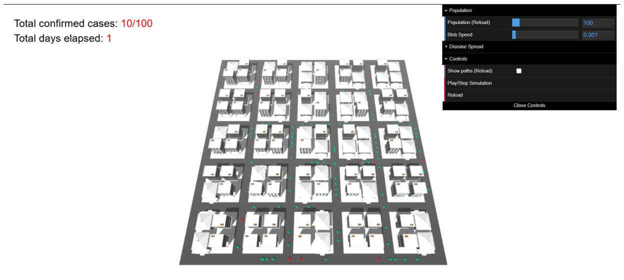

Fig. 1. Visual interface of the pandemic simulation showing agent layout and control panel.

VI. Conclusion and Future Work

This simulation offers a visually engaging and educational tool for exploring epidemic dynamics. Its accessibility via the web and its interactive design make it suitable for classrooms, awareness programs, and hobbyist experimentation. Future improvements could include:

- Adding vaccination agents or quarantine zones
- Using real-world data for parameter tuning
- Extending to 2D/3D hybrid visualizations with UI overlays

Such enhancements would further increase the simulation's realism and utility as a public education tool.

References

[1] Wilensky, U. (1999). NetLogo. Center for Connected Learning and Computer-Based Modeling, Northwestern University.
[2] Ferguson, N., et al. (2020). Report 9: Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand. Imperial College London.
[3] Eubank, S., et al. (2004). Modelling disease outbreaks in realistic urban social networks. Nature.